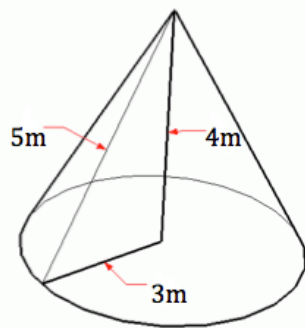


8.4/8.5 Volume and Surface Area of Cones Worksheet

MPM1D

1. Find the volume and surface area of each cone:

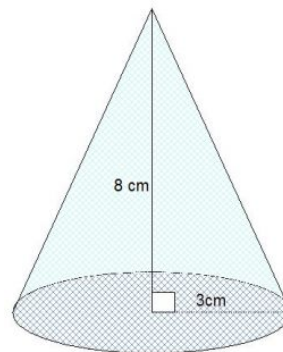
a)



Volume= _____

Surface Area= _____

b)

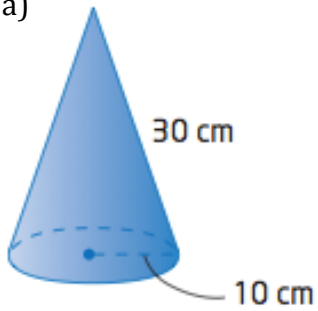


Volume= _____

Surface Area= _____

2. Find the surface area and volume of each cone. *Hint: You will have to use Pythagorean Theorem to solve for the missing measurement.*

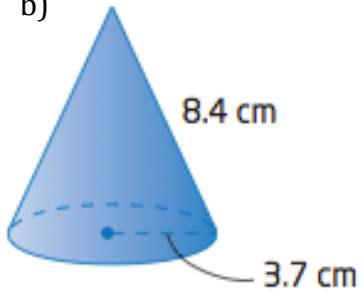
a)



Volume= _____

Surface Area= _____

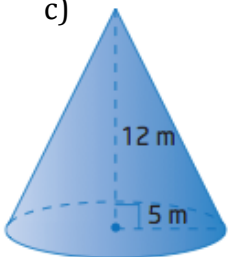
b)



Volume= _____

Surface Area= _____

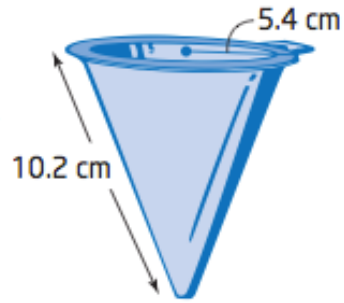
c)



Volume= _____

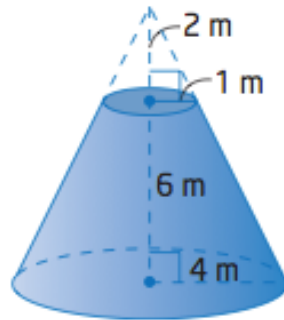
Surface Area= _____

3. Wesley uses a cone-shaped funnel to put oil in a car engine. The funnel has a radius of 5.4 cm and a slant height of 10.2 cm. How much oil can the funnel hold, to the nearest tenth of a cubic centimetre?

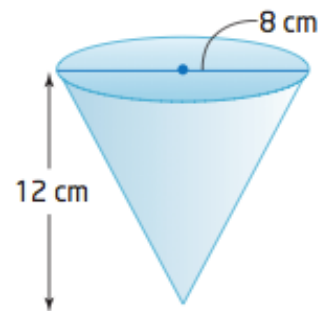


4. A cone-shaped paper cup has a volume of 67 cm^3 and a diameter of 6 cm. What is the height of the paper cup, to the nearest tenth of a centimetre?
5. A cone just fits inside a cylinder with volume 300 cm^3 . What is the volume of the cone?
6. A cone has a volume of 150 cm^3 . What is the volume of a cylinder that just holds the cone?

7. The frustum of a cone is the part that remains after the top portion has been removed by making a cut parallel to the base. Calculate the surface area of this frustum, to the nearest square metre.



8. Some paper cups are shaped like cones. How much paper, to the nearest square centimeter, is needed to make the cup?



Answers:

1) a) $V=37.7 \text{ m}^3$; $SA=75.4 \text{ m}^2$ b) $V= 75.4 \text{ cm}^3$; $SA= 108.4 \text{ cm}^2$ 2) a) $V= 2963.6 \text{ cm}^3$; $SA= 1256.7 \text{ cm}^2$
 b) $V= 107.5 \text{ cm}^3$; $SA= 140.6 \text{ cm}^2$ c) $V= 314.2 \text{ m}^3$; $SA= 282.7 \text{ m}^2$ 3) 265.7 cm^3 4) 7.1 cm 5) 100 cm^3
 6) 450 cm^3 7) 158.2 m^2 8) 158.3 cm^2